Basic Controls and Event Handling (Lessons 1-3)

This phase focuses on user interface (GUI) design and event-driven programming:

• Form Design: Use Toolbox to drag and drop controls and the Properties window to set properties.

• Interaction: Use Labels to display static text and Buttons to trigger events. Learn how to write code to handle Click events to respond to user actions.

• Data Input/Output: Use TextBoxes to receive input from the keyboard and MessageBoxes to display pop-up messages.

Logic and List Data Processing (Lessons 4-5)

These exercises enhance the ability to handle data behind the interface:

• Data Type Conversion: In a simple calculator problem, you learned how to convert a text string to a number (double) using Convert.ToDouble() to perform addition, subtraction, multiplication, and division operations.

• Selection Lists: Familiarize yourself with ListBox and ComboBox. I know how to add items to a list via code (e.g., in the Form_Load event) and retrieve the user-selected value via the .SelectedItem property.

Managing Table Data with DataGridView (Lessons 6-9)

This is the core knowledge section on displaying and manipulating complex data:

• Displaying data: Configure DataGridView to create header columns and add static data rows

• Dynamic manipulation: Add new rows to the table from data entered in a TextBox and delete the selected row using the .RemoveAt() method based on the row index (SelectedRows.Index).

• Advanced: Integrate DataGridViewComboBoxColumn to create a dropdown list directly inside a data table cell, increasing flexibility when entering data.

Time and Process Control (Lesson 10)

The final exercise introduces dynamic interface components:

• DateTimePicker: Provides a calendar interface for users to select precise dates.

• Process Simulation: Uses a Timer to trigger events periodically (e.g., every 100ms) to update the value of the ProgressBar, simulating background tasks or data loading processes.